



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - Prova scritta del 17/02/2022

PROBLEM

The random variable X has uniform distribution between $-\alpha/2$ and $\alpha/2$, where $\alpha > 0$.

Assuming that we observe N independent realizations of X :

- (1) Plot the 1st order pdf of X and then derive and plot the likelihood function $L(\alpha)$.
- (2) Derive an estimator of parameter α by applying the method of moments (MM). If possible, find its bias and mean square error (MSE) and establish if the estimator is unbiased and consistent.
- (3) Derive the maximum likelihood (ML) estimator of parameter α , its bias and mean square error (MSE), and establish if the estimator is unbiased and consistent.
- (4) Comment on the pros and cons of the MM and ML estimators.



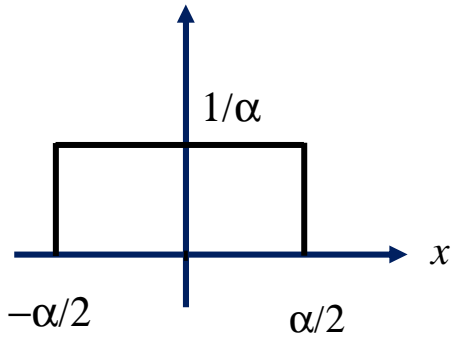
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Solution:

(1) The 1st order pdf is given by

$$f_X(x; \alpha) = \frac{1}{\alpha} \text{rect}\left(\frac{x}{\alpha}\right) = \begin{cases} 1/\alpha, & -\alpha/2 \leq x \leq \alpha/2 \\ 0, & \text{otherwise} \end{cases}$$



Since the observed data are independent and identically distributed (IID), the likelihood function (LF) is given by the product of the marginal probability density functions (pdf's):

$$\begin{aligned} L(\alpha) &= \prod_{i=1}^N f_X(x_i; \alpha) = \prod_{i=1}^N \frac{1}{\alpha} \text{rect}\left(\frac{x_i}{\alpha}\right) = \frac{1}{\alpha^N} \prod_{i=1}^N \text{rect}\left(\frac{x_i}{\alpha}\right) \\ &= \begin{cases} \frac{1}{\alpha^N}, & -\frac{\alpha}{2} \leq x_1 \leq \frac{\alpha}{2}, -\frac{\alpha}{2} \leq x_2 \leq \frac{\alpha}{2}, \dots, -\frac{\alpha}{2} \leq x_N \leq \frac{\alpha}{2} \\ 0, & \text{otherwise} \end{cases} \\ &= \begin{cases} \frac{1}{\alpha^N}, & |x_1| \leq \frac{\alpha}{2}, |x_2| \leq \frac{\alpha}{2}, \dots, |x_N| \leq \frac{\alpha}{2} \\ 0, & \text{otherwise} \end{cases} \\ &= \begin{cases} \frac{1}{\alpha^N}, & \max\{|x_i|\} \leq \frac{\alpha}{2} \\ 0, & \text{otherwise} \end{cases} \end{aligned}$$

The LF can also be expressed as:

$$L(\alpha) = \frac{1}{\alpha^N} u\left(\frac{\alpha}{2} - \max\{|x_i|\}\right) = \frac{1}{\alpha^N} u\left(\alpha - 2 \max\{|x_i|\}\right) \quad (*)$$



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(2) The mean of X is 0, so we cannot use it to estimate α ; we should resort to the variance (or the power) to find a Method of Moments (MM) estimator:

$$m_{1X} = E\{X\} = 0 \quad (\text{the mean value is the symmetry point of the pdf})$$

$$\mu_2 = E\{(X - m_1)^2\} = E\{X^2\} = \frac{\alpha^2}{12} \Rightarrow \alpha = \sqrt{12\mu_2} \Rightarrow \hat{\alpha}_{MM1}(\mathbf{x}) = \sqrt{12\hat{\mu}_2} = \sqrt{\frac{12}{N} \sum_{i=1}^N X_i^2}$$

We can immediately say that the estimator is non-linear and biased, but consistent. However, it is difficult to find the analytic expressions of bias and MSE for finite sample size, i.e. as a function of N .

Another possibility to find an estimator by using the MM is based on the observation that $Y \triangleq |X|$ is a r.v. uniformly distributed between 0 and $\alpha/2$, so we have $E\{Y\} = E\{|X|\} = \alpha/4$. Consequently, a possible estimator of α based on the MM can be derived as follows:

$$\alpha = 4m_{1Y} \Rightarrow \hat{\theta}_{MM2}(\mathbf{x}) = 4\hat{m}_{1Y} = \frac{4}{N} \sum_{i=1}^N Y_i = \frac{4}{N} \sum_{i=1}^N |X_i|$$

$$E\{\hat{\theta}_{MM2}(\mathbf{x})\} = E\left\{\frac{4}{N} \sum_{i=1}^N |X_i|\right\} = \frac{4}{N} \sum_{i=1}^N E\{|X_i|\} = \frac{4}{N} \sum_{i=1}^N \frac{\alpha}{4} = \alpha$$

$$MSE\{\hat{\theta}_{MM2}(\mathbf{x})\} = \text{var}\{\hat{\theta}_{MM2}(\mathbf{x})\} = \text{var}\left\{\frac{4}{N} \sum_{i=1}^N |X_i|\right\} = \frac{16}{N^2} \sum_{i=1}^N \text{var}\{|X_i|\} = \frac{16}{N^2} \cdot N \cdot \frac{(\alpha/2)^2}{12} = \frac{\alpha^2}{3N}$$

This MM estimator, called MM2, is unbiased and consistent.

(3) In this case, the likelihood function is not continuous and differentiable in the definition interval of the parameter α , that is $\alpha \in (0, +\infty)$. Hence, the ML estimate cannot be obtained by solving the likelihood equation, but it must be obtained by looking directly for the value of α for which $L(\alpha)$ is maximum. In our case, from equation (*) we immediately obtain:

$$\hat{\alpha}_{ML}(\mathbf{x}) = \arg \max_{\alpha} L(\alpha) = 2 \max\{|X_1|, |X_2|, \dots, |X_N|\} = \max\{Z_1, Z_2, \dots, Z_N\}$$

Where we defined $Z_i \triangleq 2|X_i| \in U(0, \alpha)$, $i = 1, 2, \dots, N$



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Following the usual derivation, (see course notes Ch. 4 pages 52-66) we easily find:

$$f_{\hat{\alpha}_{ML}}(a) = \frac{N}{\alpha} \left(\frac{a}{\alpha} \right)^{N-1} \text{rect} \left(\frac{a - \alpha/2}{\alpha} \right)$$

$$E\{\hat{\alpha}_{ML}\} = \frac{N\alpha}{N+1}, \quad E\{\hat{\alpha}_{ML}^2\} = \frac{N\alpha^2}{N+2} \Rightarrow \text{var}\{\hat{\alpha}_{ML}\} = E\{\hat{\alpha}_{ML}^2\} - (E\{\hat{\alpha}_{ML}\})^2 = \frac{N\alpha^2}{(N+1)^2(N+2)}$$

$$b\{\hat{\alpha}_{ML}\} = E\{\alpha - \hat{\alpha}_{ML}\} = \alpha - E\{\hat{\alpha}_{ML}\} = \frac{\alpha}{N+1} \xrightarrow{N \rightarrow \infty} 0$$

$$MSE\{\hat{\alpha}_{ML}(\mathbf{x})\} = (b\{\hat{\alpha}_{ML}\})^2 + \text{var}\{\hat{\alpha}_{ML}\} = \frac{2\alpha^2}{(N+1)(N+2)} \xrightarrow{N \rightarrow \infty} 0$$

Hence, the ML estimator is biased but asymptotically unbiased and consistent.

$$(4) \quad \hat{\alpha}_{MM1}(\mathbf{x}) = \sqrt{\frac{12}{N} \sum_{i=1}^N X_i^2}, \quad \hat{\alpha}_{MM2}(\mathbf{x}) = \frac{4}{N} \sum_{i=1}^N |X_i|, \quad \hat{\alpha}_{ML}(\mathbf{x}) = 2 \max\{|X_1|, |X_2|, \dots, |X_N|\}$$

Performance comparison:

The MSE of the ML estimator goes to zero with the law $1/N^2$. Instead, the MSE of the MM2 estimator goes to zero with the law $1/N$. Hence, the ML estimator is much more efficient. To evaluate the performance of the MM1 estimator we have to resort to the Monte Carlo simulation. The MM1 estimator has the same asymptotical behavior as the MM2 estimator.

Comparison in terms of computational complexity:

The ML estimator is **non-linear**, it requires calculation of the maximum of the absolute values of a set of N data (data ordering). The MM1 and MM2 estimators are **non-linear**, but a linear function of the squares or of the absolute values of the data samples. Hence, both of them are much more **computationally efficient** than the ML estimator.

The choice between them depends on the constraints imposed by the specific application: if estimation accuracy is the most important issue, then we should use the ML estimate; if instead it is more important to keep low the computational complexity, we should use the MM1 or MM2 estimator.